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Rethinking Representativeness and Diversity in Dynamic Data Selection

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Abstract

Dynamic data selection accelerates training by sampling a changing subset of the dataset while preserving accuracy. We rethink two core notions underlying sample evaluation: representativeness and diversity. Instead of local geometric centrality, we define representativeness as coverage of dataset-level common or high-frequency feature factors. Instead of within-subset dispersion, we define diversity at the process level, requiring the selection trajectory to gradually include complementary rare factors over training. Based on this view, we propose a dynamic selection framework with three components. First, we score representativeness in a plug-in feature space to prioritize samples covering frequent factors. We instantiate this with a sparse autoencoder trained on the target dataset, using sparse unit activations to summarize both individual samples and dataset-wide factor statistics. Second, we realize process-level diversity by combining rare-factor sampling with a Usage-Frequency Penalty that promotes sample rotation, provably discourages monopoly, and reduces gradient bias. Third, we couple the two-dimensional scoring with a smooth scheduler that transitions selection from core-pattern consolidation to rare-factor exploration, without extra gradients, influence estimates, or second-order computations on the training model. Extensive experiments on five benchmarks across vision and text tasks demonstrate improved accuracy–efficiency trade-offs across models. Our method matches or exceeds full-data accuracy with over 2× training acceleration. *Code will be released.*

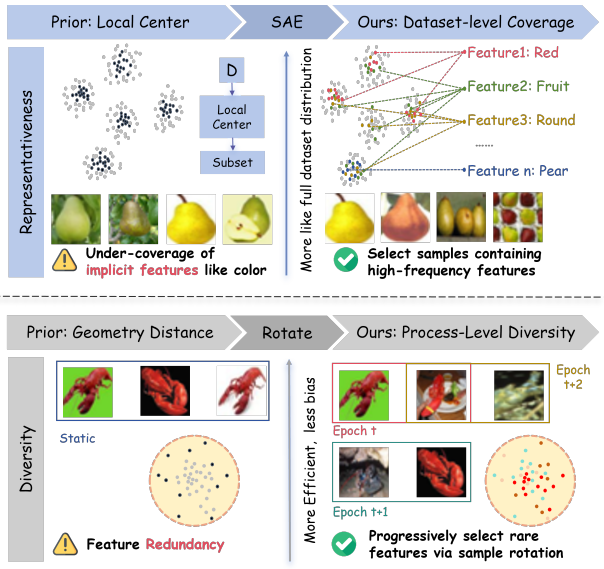

Figure 1. **Conceptual comparison with prior data selection methods.** Previous methods relying on geometry-based metrics may overemphasize local centrality and miss implicit feature factors. Our framework measures representativeness as dataset-level coverage of common or high-frequency factors, and enforces process-level diversity by promoting sample rotation across epochs rather than optimizing a single static subset.

1. Introduction

The remarkable performance of deep learning models often comes at the cost of prohibitive computation overhead from training on massive datasets (Liu et al., 2024; Touvron et al., 2023; 2021). Data selection aims to extract a small yet high-value subset from a large corpus, thereby reducing cost and wall-clock time while maintaining or even improving accuracy (Albalak et al., 2024).

Early work predominantly adopted static data selection strategies (Xie et al., 2023; Har-Peled & Mazumdar, 2004; Sener & Savarese, 2018; Mirzasoleiman et al., 2020), i.e., choosing a fixed subset before training that balances representativeness and diversity (Yang et al., 2025a; Agarwal et al., 2020; Ge et al., 2024; Yu et al., 2025; Du et al., 2025). Its fundamental limitation is a mismatch between a single, static view of the data distribution and the model's evolving capacity and learning needs across training. In practice, to preserve final accuracy, the subset size usually cannot be reduced aggressively (e.g., performance may degrade significantly unless about 70% of the data is retained (Tan et al., 2023; Xia et al., 2023; Yang et al., 2025a; Mirzasoleiman

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